

! The programme uses to find midpoint between two point.

Define a class named **Point** to model a coordinate of point. Use inline style to define the class. Follow the requirements below to create your program:

1. Define a constructor that accepts two parameters to set the attributes x and y, respectively. Declare this constructor such that it can also serves as a default constructor (or default arguments constructor). In this case, it will model the coordinate (0,0).
2. Define overloaded mutators function name setX and setY to set the attribute of x and y.
3. Define accessor function named getX and getY to get the attribute of x and y.
4. Define an overloaded operator that will return another Point object. The overloaded operator used for summing up two coordinates. For an example , coordinate 1 (5,4) , coordinate 2(3,5).

coordinate 1 + coordinate 2 = (8,9)

5. Define a method name toString() that returns a line symbolic of type string. This method will convert the x attributes and y attributes to the format of (x,y)
6. Define a copy constructor to copy the attributes of one object to another object.
7. Declare a friend class name Calculation.

Define another class named Calculation to model all coordinate of point and do calculation. (TIPS , attributes got 2 , one is num , one is dynamic allocate memory)

1. Define a constructor that accepts one parameter to set the number of point and construct an array for Point object.
2. Define a destructor and delete to dynamic allocate memory.
3. Define a accessor method name getCoordinate to get the dynamic stored value in an array
4. Define a function named calculateMidpoint that two int number which user input to choose wheter which one point to be used for calculation midpoint. Use operator+ function that declare at the class Point to do the operation sum up. In this function , also invoke the copy constructor to move the result from operation + of 2 Point objects to a new Point object. This function will then return a Point object.

5. Define a function named toString to return a string data type. The string return should consist of all point that users input before and then also print the midpoint of 2 points user selected.
6. Define a function name input to let user assign value to the array element(Point's object).

In the main function, invoke the functions that define before and make sure the output will print as below example output:

```
How many points to record : 3

Point 1
Enter X-coordinates: 1
Enter Y-coordinates: 5

Point 2
Enter X-coordinates: 3
Enter Y-coordinates: 7

Point 3
Enter X-coordinates: 9
Enter Y-coordinates: -3

Which 2 point used to calculate the midpoint? : 1 3

Point 1 coordinate:(1.000000,5.000000)
Point 2 coordinate:(3.000000,7.000000)
Point 3 coordinate:(9.000000,-3.000000)

Midpoint of the choosen 2 point (1.000000,5.000000) and (9.000000,-3.000000) :(5.000000,1.000000)
Press any key to continue . . . |
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